

Exp:6

```
(sandhiya@LAPTOP-OBAR95LN)~/mnt/c/Users/Sandhiya
$ nano exp6.sh

(sandhiya@LAPTOP-OBAR95LN)~/mnt/c/Users/Sandhiya
$ chmod +x exp6.sh

(sandhiya@LAPTOP-OBAR95LN)~/mnt/c/Users/Sandhiya
$ ./exp6.sh or bash exp6.sh
Entering Critical Section
4
Leaving Critical Section
```

EX-7

```
(admin@DESKTOP-THKRVE0)~/mnt/c/Users/ADMIN
$ nano exp7.sh

(admin@DESKTOP-THKRVE0)~/mnt/c/Users/ADMIN
$ chmod +x exp7.sh
chmod: changing permissions of 'exp7.sh': Operation not permitted

(admin@DESKTOP-THKRVE0)~/mnt/c/Users/ADMIN
$ ./exp7.sh or bash exp7.sh
Enter Number of Processes:
4
Enter Safe Sequence (space separated):
2 5 7 3
Safe Sequence is:
P2 P5 P7 P3
System is in Safe State
```

EX-8

```
(admin@DESKTOP-THKRVE0)~/mnt/c/Users/ADMIN
$ nano exp8.sh

(admin@DESKTOP-THKRVE0)~/mnt/c/Users/ADMIN
$ chmod +x exp8.sh
chmod: changing permissions of 'exp8.sh': Operation not permitted

(admin@DESKTOP-THKRVE0)~/mnt/c/Users/ADMIN
$ ./exp8.sh or bash exp8.sh
Enter number of processes:
7 9 9 2
Enter deadlocked process numbers (if any):
223
Deadlocked Processes:
223
```

EX 9:

```

(admin@DESKTOP-THKRVE0)-[/mnt/c/Users/ADMIN]
$ nano exp9.sh

(admin@DESKTOP-THKRVE0)-[/mnt/c/Users/ADMIN]
$ chmod +x exp9.sh

(admin@DESKTOP-THKRVE0)-[/mnt/c/Users/ADMIN]
$ ./exp9.sh or bash exp9.sh
Thread 1 : 1
Thread 2 : 1
Thread 1 : 2
Thread 2 : 2
Thread 1 : 3
Thread 2 : 3
Thread 1 : 4
Thread 2 : 4
Thread 1 : 5
Thread 2 : 5
All Threads Completed

```

Ex 10

```

(sandhiya@LAPTOP-OBAR95LN)-[/mnt/c/Users/Sandhiya]
$ chmod +x exp10.sh

(sandhiya@LAPTOP-OBAR95LN)-[/mnt/c/Users/Sandhiya]
$ ./exp10.sh or bash exp10.sh
Enter Page Size:
4
Enter Number of Pages:
4
Enter Frame Number for Page 0:
9
Enter Frame Number for Page 1:
7
Enter Frame Number for Page 2:
3
Enter Frame Number for Page 3:
5
Enter Logical Address:
2
Page Number : 0
Offset : 2
Frame Number : 9
Physical Address : 38

```

EXp 11A

```

(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ chmod +x exp11a.sh

(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ ./exp11a.sh
First Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using First Fit

```

EXP 11B

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ chmod +x exp11b.sh
```

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ ./exp11b.sh
```

Best Fit Memory Allocation Demonstration

Memory Blocks: 100 500 200 300 600

Processes: 212 417 112 426

Allocation Performed Using Best Fit

#### Exp 11(C)

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ chmod +x exp11c.sh
```

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ ./exp11c.sh
```

Worst Fit Memory Allocation Demonstration

Memory Blocks: 100 500 200 300 600

Processes: 212 417 112 426

Allocation Performed Using Worst Fit

#### EXP 12(A)

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ chmod +x exp12a.sh
```

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ ./exp12a.sh
```

FIFO Page Replacement Demonstration

Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2

Frames: 3

FIFO Algorithm Executed

#### Exp12)B)

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
└─$ chmod +x exp12b.sh

(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
└─$ ./exp12b.sh
LRU Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
LRU Algorithm Executed
```

EXP 12)C)

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
└─$ chmod +x exp12c.sh

(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
└─$ ./exp12c.sh
Optimal Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
Optimal Algorithm Executed
```

EXP 13(a)

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
└─$ chmod +x exp13a.sh

(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
└─$ ./exp13a.sh
Enter Register Number:
101
Enter Name:
Ram
Contents of File
101 Ram
```

EXP 13 B:

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ chmod +x exp13b.sh
```

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ ./exp13b.sh
```

Enter Record:

102 Shyam

Random Access Record

102 Shyam

EXP13)C)

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ chmod +x exp13c.sh
```

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ ./exp13c.sh
```

Enter Student Records

Enter Register Number to Search:

102

102 Kumar

EXP 14)A)

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ chmod +x exp14a.sh
```

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]  
$ ./exp14a.sh
```

Enter Starting Block:

10

Enter File Length:

5

Allocated Blocks:

10 11 12 13 14

EXP 14)B)

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ chmod +x exp14b.sh
```

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ ./exp14b.sh
```

Enter Index Block:

50

Enter Number of Blocks:

4

Enter Block Numbers:

12

19

22

45

Index Block : 50

Allocated Blocks : 12 19 22 45

EXP 14(C)

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ chmod +x exp14c.sh
```

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ ./exp14c.sh
```

Enter Number of Blocks:

4

Enter Block Numbers:

10

15

22

38

Linked Allocation:

10 --> 15 --> 22 --> 38 --> NULL

EX 15)A

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ chmod +x exp15a.sh

(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ ./exp15a.sh
Total Head Movement = 640
```

#### EXP 15 B

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ chmod +x exp15b.sh

(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ ./exp15b.sh
SSTF Disk Scheduling Demonstration
Request Queue : 98 183 37 122 14 124 65 67
Initial Head Position : 53
Total Head Movement calculated using SSTF.
```

#### EXP 15(C)

```
(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ chmod +x exp15c.sh

(jetasri@Desktop-OH47NGA)-[/mnt/c/Users/Jetasri]
$ ./exp15c.sh
SCAN Disk Scheduling
Head moves in one direction and then reverses.
```

#### EXP 15(D)

```
(jetasri@Desktop-OH47NGA) - [/mnt/c/Users/Jetasri]  
$ chmod +x exp15d.sh
```

```
(jetasri@Desktop-OH47NGA) - [/mnt/c/Users/Jetasri]  
$ ./exp15d.sh
```

C-SCAN Disk Scheduling

Head moves in one direction only.

### Exp6:

```
Parent Process Entering Critical Section  
Parent Process Leaving Critical Section  
Child Process Entering Critical Section  
Child Process Leaving Critical Section
```

```
=== Code Execution Successful ===
```

### Exp 7:

#### Output

```
Enter Number of Processes: 2  
Enter Number of Resources: 2
```

```
Enter Allocation Matrix:
```

```
10  
20  
30  
40
```

```
Enter Maximum Matrix:
```

```
15  
25  
35  
45
```

```
Enter Available Resources:
```

```
5 5
```

```
System is in Safe State
```

```
Safe Sequence: P0 P1
```

```
=== Code Execution Successful ===
```

### Exp 8:

```
Console I/O AI Agent
Enter Number of Processes: 2
Enter Number of Resource Types: 2

Enter Allocation Matrix:
10
20
50
80

Enter Request Matrix:
4
20
30
50

Enter Available Resources:
15
25

Deadlocked Processes:
P1
```

EXP 9:

```
Console I/O AI Agent
Thread Executing : 1
Thread Executing : 1
Thread Executing : 2
Thread Executing : 2
Thread Executing : 3
Thread Executing : 3
Thread Executing : 4
Thread Executing : 4
Thread Executing : 5
Thread Executing : 5
All Threads Completed
```

**EX 10:**

```
Enter Page Size: 5
Enter Number of Pages: 3
Enter Frame Numbers for Each Page:
Page 0 -> Frame: 1
Page 1 -> Frame: 2
Page 2 -> Frame: 3
Enter Logical Address: 4

Page Number : 0
Offset : 4
Frame Number : 1
Physical Address : 9

=== Code Execution Successful ===
```

**EX 11:**

**PROGRAM A: FIRST FIT(c program)**

```
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
4
5
Enter Process Sizes:
6
7

Process No   Process Size   Block No
1           6         Not Allocated
2           7         Not Allocated

=== Code Execution Successful ===
```

11)PROGRAM B: BEST FIT

Enter Number of Blocks: 3

Enter Number of Processes: 2

Enter Block Sizes:

2

6

7

Enter Process Sizes:

8

9

| Process No | Process Size | Block No |
|------------|--------------|----------|
|------------|--------------|----------|

|   |   |               |
|---|---|---------------|
| 1 | 8 | Not Allocated |
|---|---|---------------|

|   |   |               |
|---|---|---------------|
| 2 | 9 | Not Allocated |
|---|---|---------------|

11)PROGRAM C: WORST FIT

```
Enter Number of Blocks: 3
Enter Number of Processes: 2
Enter Block Sizes:
1
7
8
Enter Process Sizes:
9
0

Process No   Process Size   Block No
1           9         Not Allocated
2           0           3
```

EXP 12(A)

```
Enter Number of Pages: 2
Enter Reference String:
3
4
Enter Number of Frames: 5
Total Page Faults = 2

=== Code Execution Successful ===
```

EXP 12(B)

```
Enter Number of Pages: 3
Enter Reference String:
2
3
1
Enter Number of Frames: 2
Total Page Faults = 3

=== Code Execution Successful ===
```

#### 12)PROGRAM C: OPTIMAL PAGE REPLACEMENT

```
Enter Number of Pages: 2
Enter Reference String:
1
2
Enter Number of Frames: 3
Total Page Faults = 2

=== Code Execution Successful ===
```

```
Enter Register Number: 101  
Enter Name: Ram
```

```
Record Details  
Register Number : 101  
Name : Ram
```

### 13)PROGRAM B: DIRECT (RANDOM) FILE ORGANIZATION

```
Enter Register Number: 102  
Enter Name: Shyam
```

```
Record Found  
Reg No : 102  
Name : Shyam
```

### 13)C)INDEXED FILE ORGANIZATION

```
Enter 3 Student Records
101 Alice
102 Bob
103 Charlie
Enter Register Number to Search: 102

Record Found
Reg No : 102
Name : Bob
```

#### 14)A)PROGRAM A: SEQUENTIAL FILE ALLOCATION

```
Enter Starting Block: 4
Enter File Length (Number of Blocks): 5

Allocated Blocks:
4 5 6 7 8

=== Code Execution Successful ===
```

#### 14)PROGRAM B: INDEXED FILE ALLOCATION

```
Enter Index Block: 2
Enter Number of Blocks: 1
Enter Block Numbers:
3
```

```
Index Block : 2
Allocated Blocks : 3
```

== Code Execution Successful ==

#### 14)PROGRAM C: LINKED FILE ALLOCATION

```
Enter Number of Blocks: 5
Enter Block Numbers:
3
4
2
1
0

Linked Allocation:
3 --> 4 --> 2 --> 1 --> 0 --> NULL
```

#### EX 15)PROGRAM A: FCFS DISK SCHEDULING

```
Enter Number of Requests: 3
Enter Request Queue:
2
1
7
Enter Initial Head Position: 5
Total Head Movement = 10

=== Code Execution Successful ===
```

#### 15)PROGRAM B: SSTF DISK SCHEDULING

```
Enter Number of Requests: 4
Enter Request Queue:
6
7
8
9
Enter Initial Head Position: 2
Total Head Movement = 7

=== Code Execution Successful ===
```

#### 15)PROGRAM C: SCAN DISK SCHEDULING

```
SCAN Disk Scheduling
```

```
Initial Head Position : 53
```

```
Head moves towards higher cylinders,  
then reverses direction.
```

```
=== Code Execution Successful ===
```

#### 15)PROGRAM D: C-SCAN DISK SCHEDULING

```
C-SCAN Disk Scheduling
```

```
Initial Head Position : 53
```

```
Head moves in one direction.
```

```
After reaching the end, it returns to the beginning.
```

```
=== Code Execution Successful ===
```